FlowchartFun

Draw arrows

OCR GCSE Computer Science (J277) flowchart practice.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.

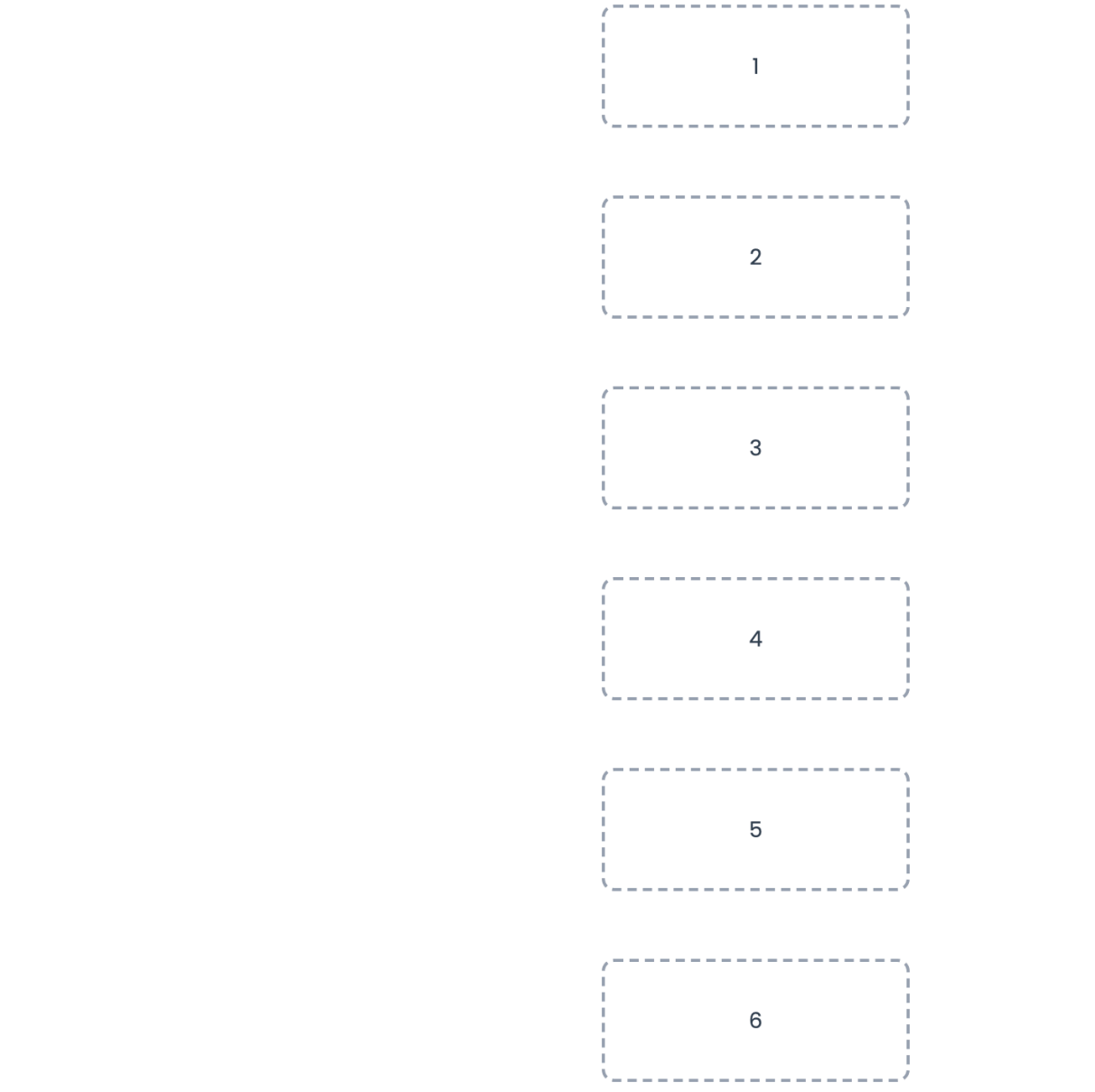
Name: _____ Class: _____

Level 1: Flowchart Basics

Easy · Draw Arrows

Create a simple flowchart that calculates the sum of two numbers. Drag each symbol into the correct place.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.

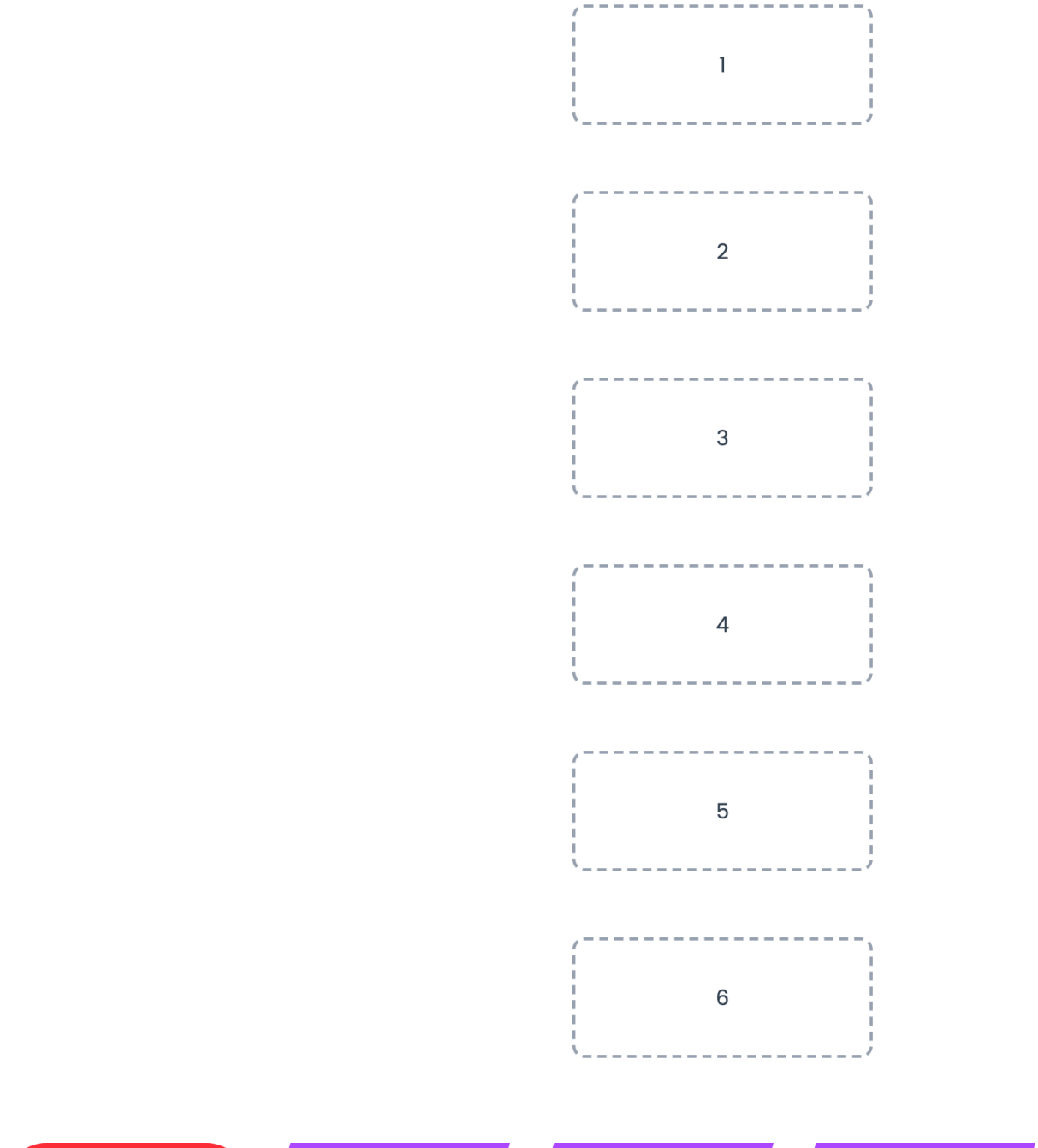


Level 2: Input, Process, Output

Easy · Draw Arrows

Make a flowchart that finds the area of a rectangle. Watch out: one symbol is not needed!

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.



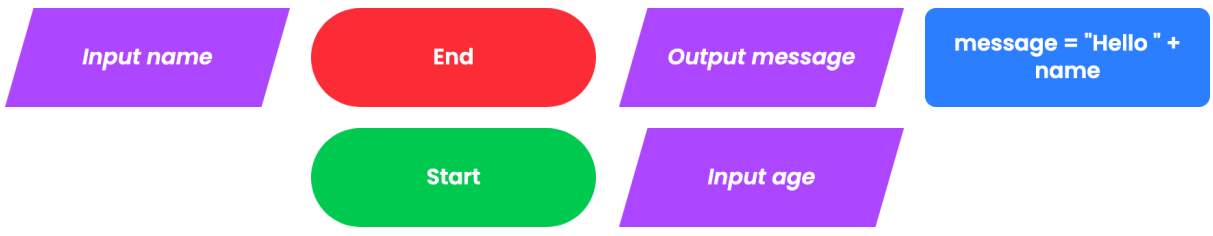
Level 3: Hello, Name

Easy · Draw Arrows

Ask for a name and greet the user. Not every symbol is needed.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.





Level 4: Average of Two

Easy · Draw Arrows

Input two numbers and output their average. Remember the brackets!

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.

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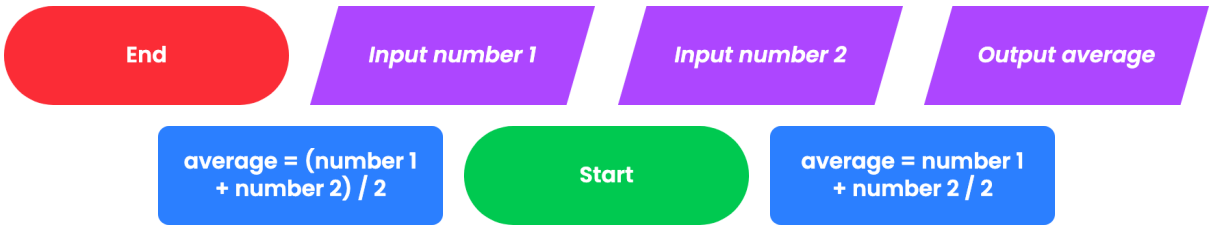
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Level 5: Making a Decision

Easy · Draw Arrows

Output whether a person is an adult or a child. Remember: diamonds are decisions, with a Yes path and a No path.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.

1

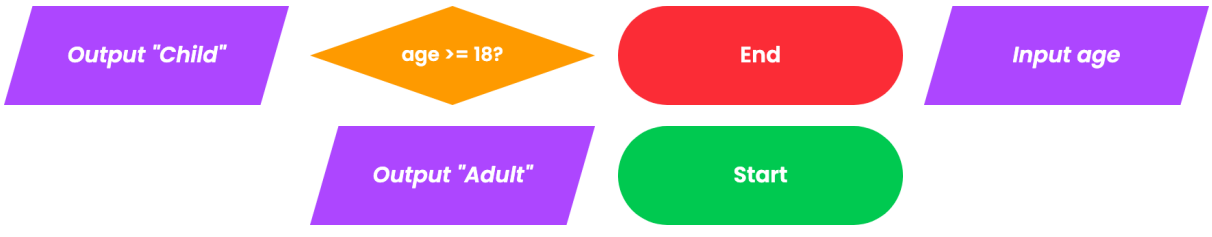
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Level 6: Odd or Even

Easy · Draw Arrows

Use MOD (the remainder after division) to decide whether a number is even or odd.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.

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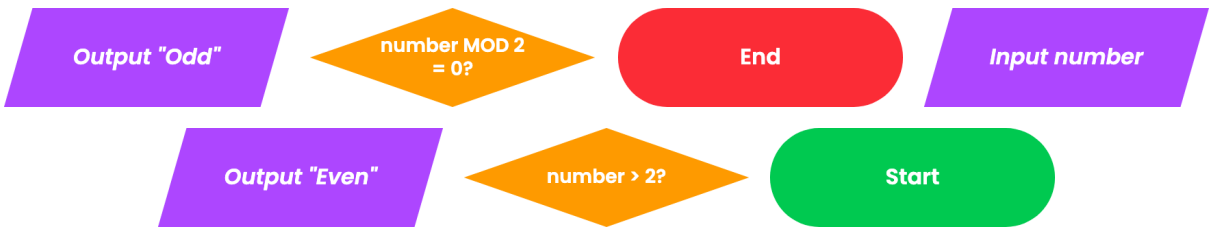
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Level 7: Pass or Fail

Easy · Draw Arrows

A mark of 40 or more is a pass. Anything else is a fail.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.

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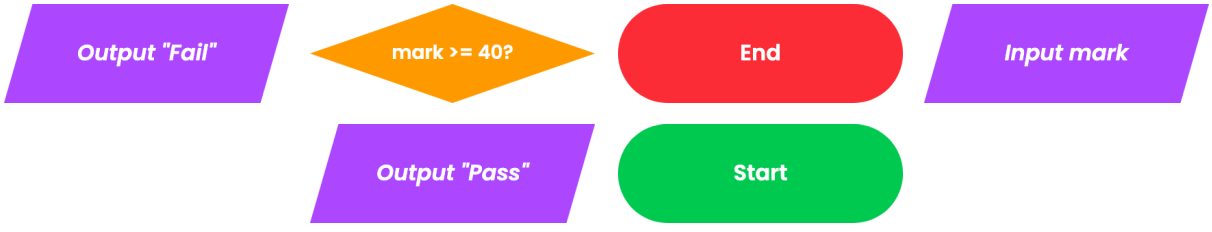
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Level 8: Discount

Easy · Draw Arrows

Shoppers get 10% off if they spend more than £100. Everyone sees a price at the end. Notice that one path skips a step.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.

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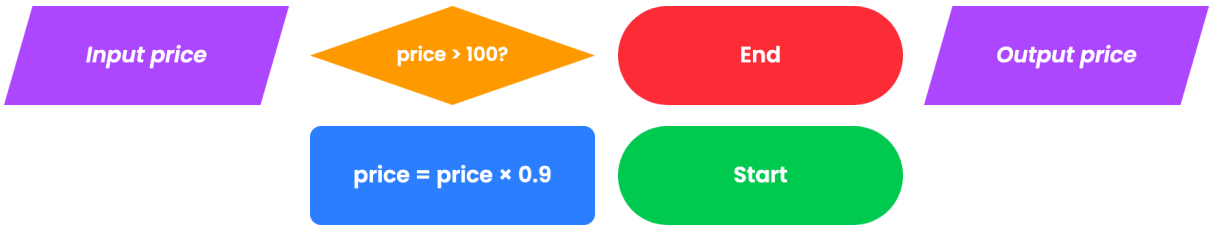
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Level 9: Cinema Ticket

Easy · Draw Arrows

Under 16s pay the child price. Everyone else pays the adult price.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.

1

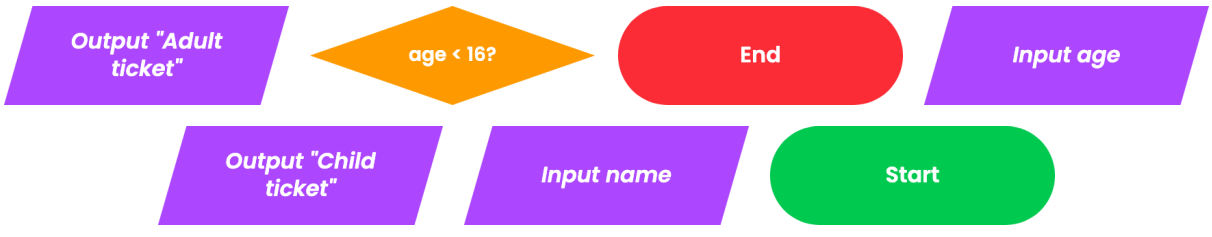
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Level 10: Login

Easy · Draw Arrows

Ask for a username and password, then decide whether to let the user in.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.

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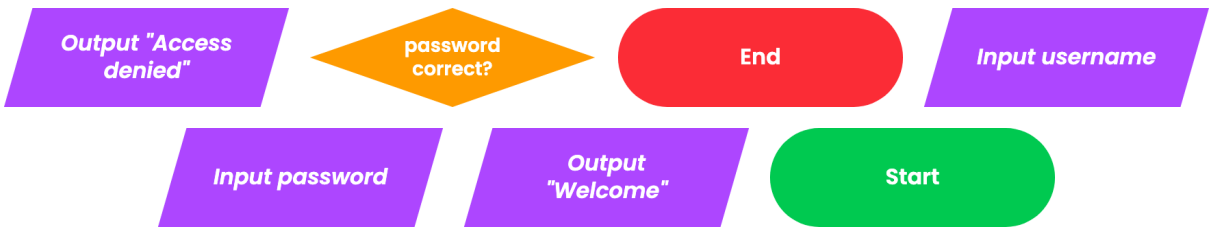
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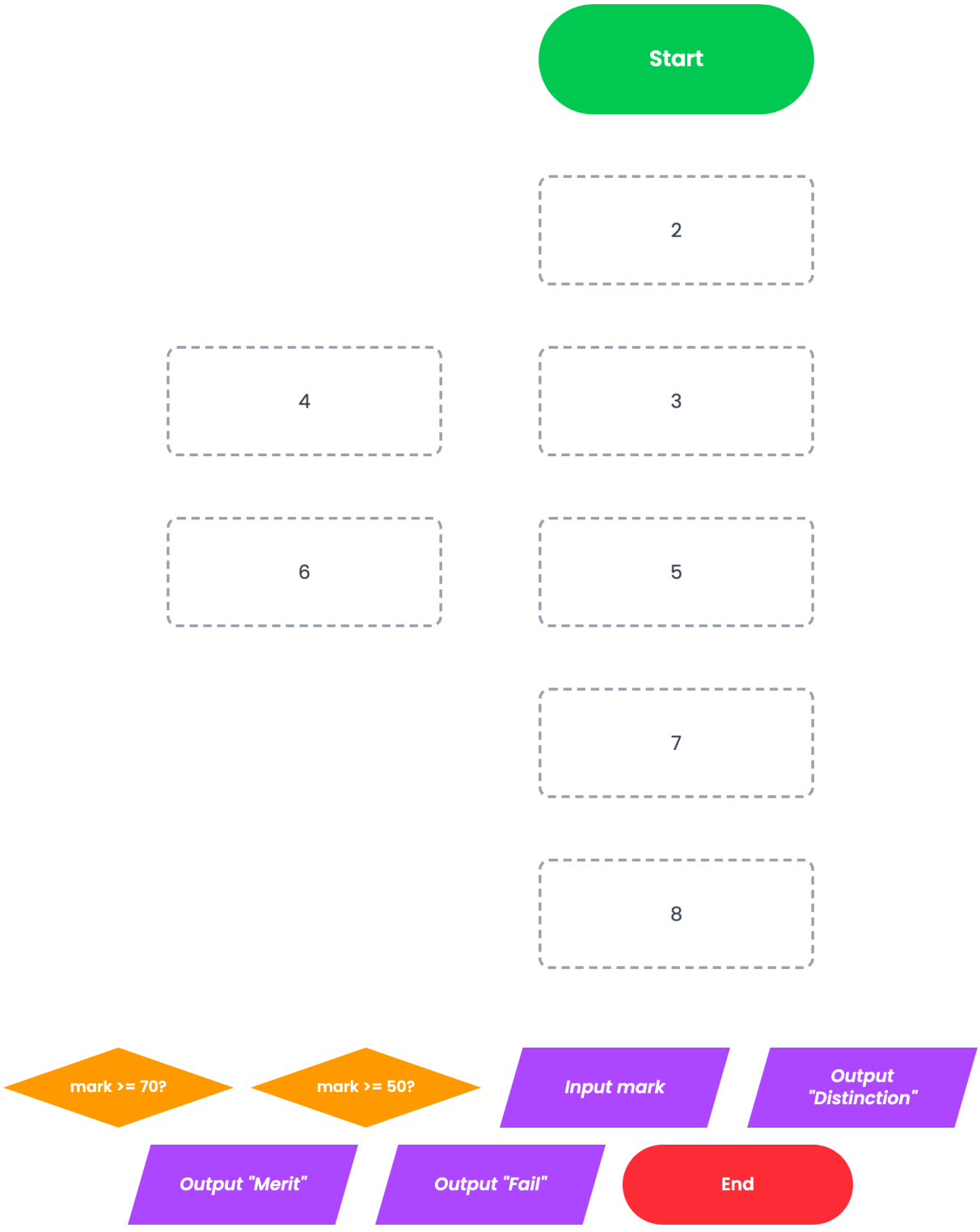


Level 11: Grade Boundaries

Medium · Draw Arrows

Output Distinction for 70 or more, Merit for 50 or more, otherwise Fail. Start is already placed for you.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.



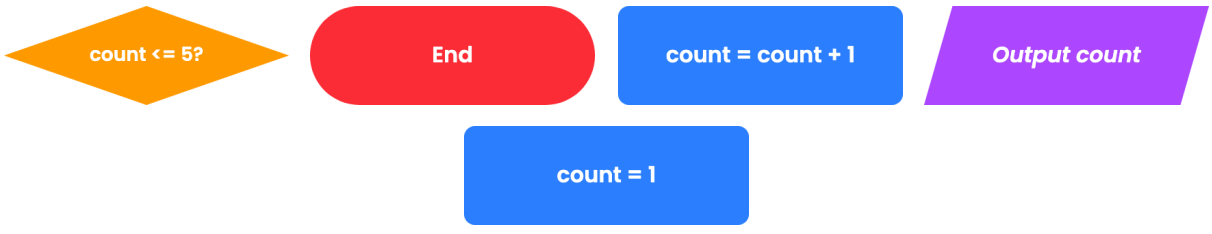
Level 12: Count to Five

Medium · Draw Arrows

Output the numbers 1 to 5. The arrow that goes back up is a loop. Start is already placed for you.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.



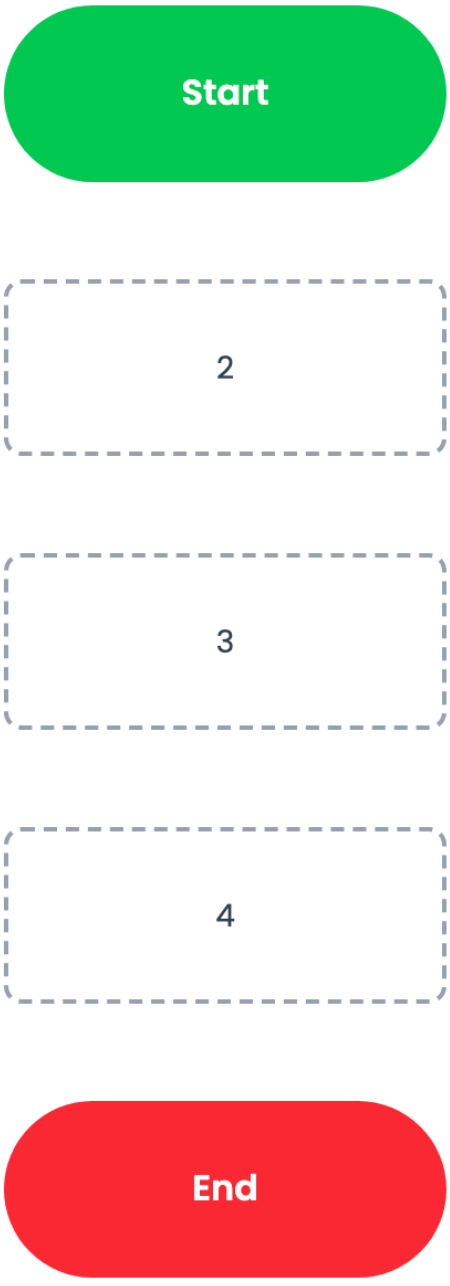


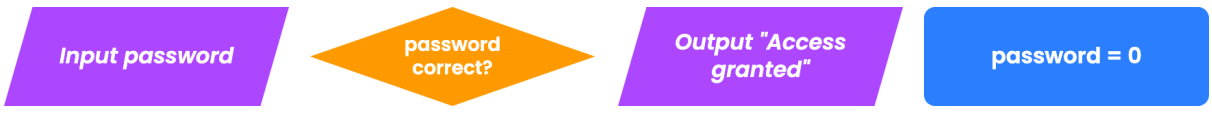
Level 13: Password Check

Medium · Draw Arrows

Keep asking for the password until it is correct, then grant access. Start and End are placed for you.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.



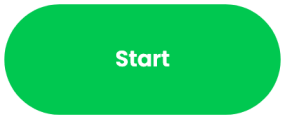


Level 14: Running Total

Medium · Draw Arrows

Input three numbers and output their total. This one combines a counter, a total and a loop.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.



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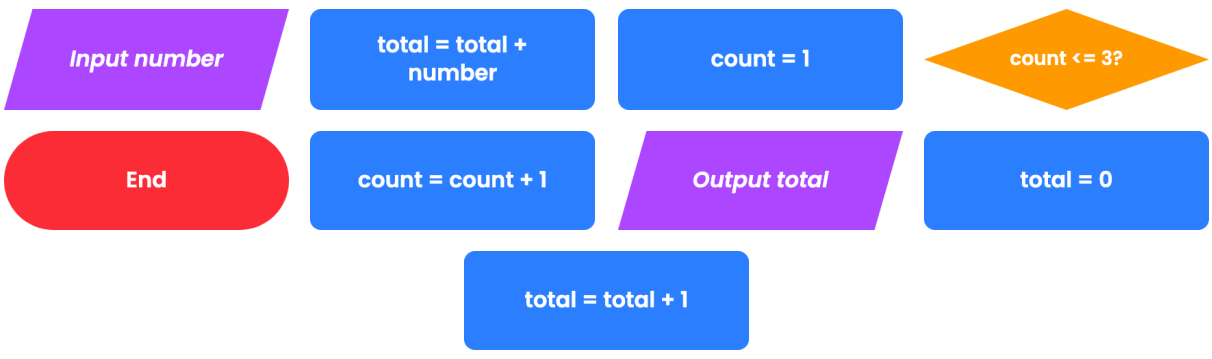
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Level 15: Larger Number

Medium · Draw Arrows

Input two numbers and output the larger one.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.

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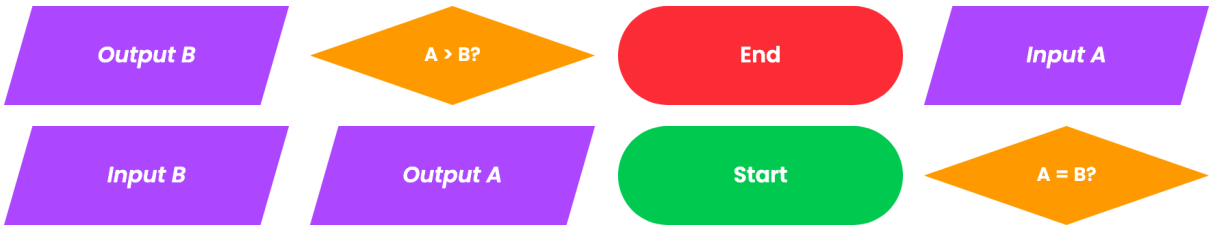
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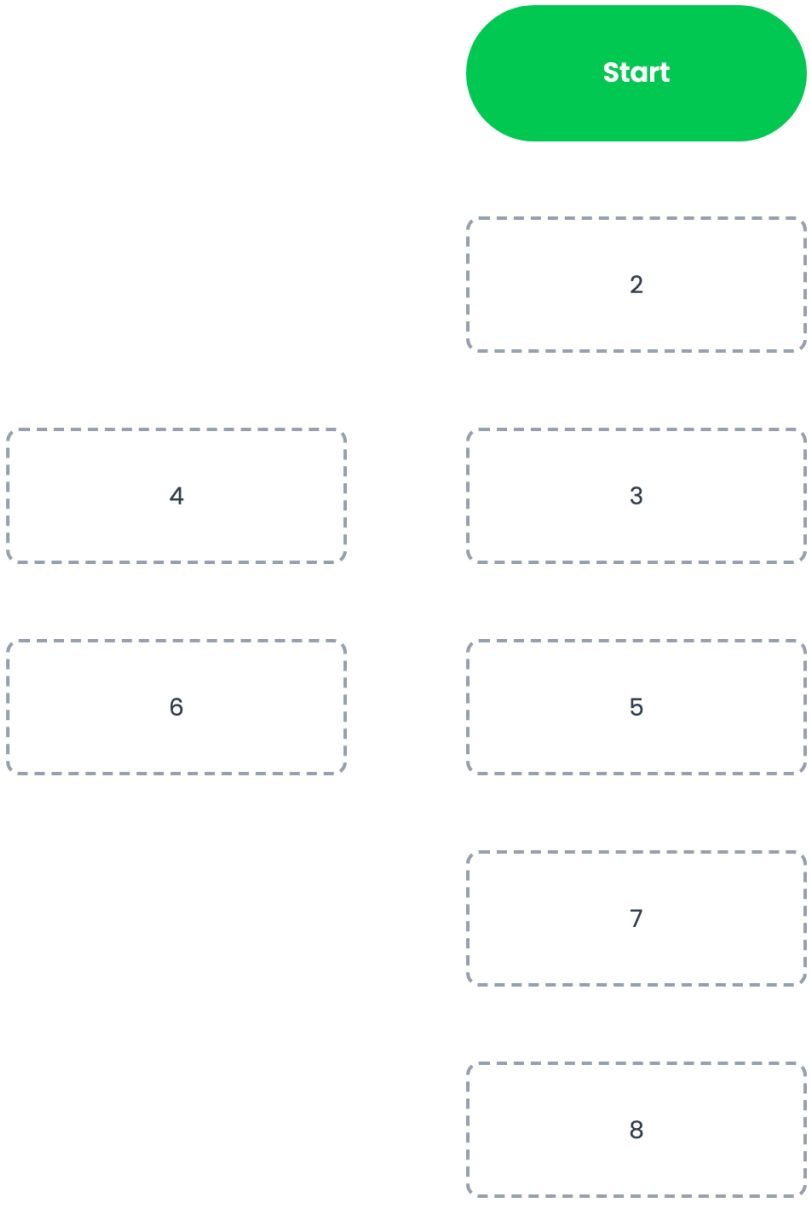


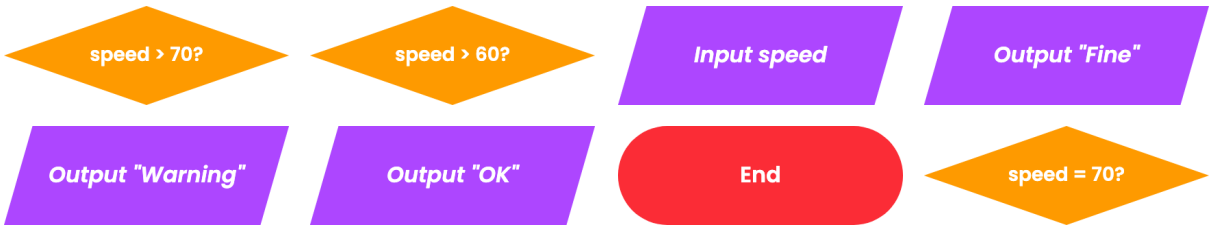
Level 16: Speed Camera

Medium · Draw Arrows

Over 70 is a fine. Over 60 is a warning. Anything else is fine to drive on. Start is already placed for you.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.





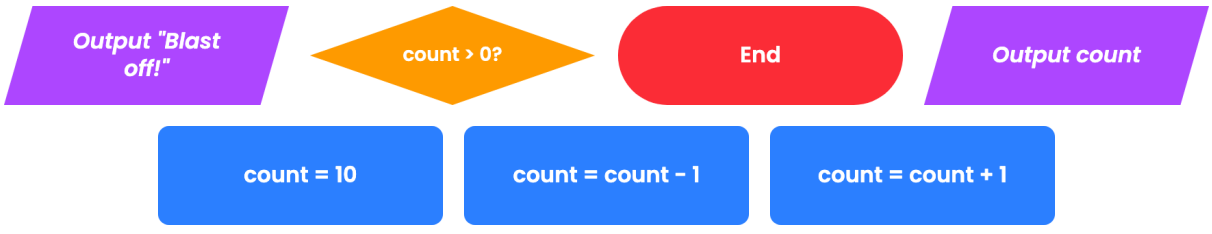
Level 17: Countdown

Medium · Draw Arrows

Count down from 10 and finish with a launch message. Start is already placed for you.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.



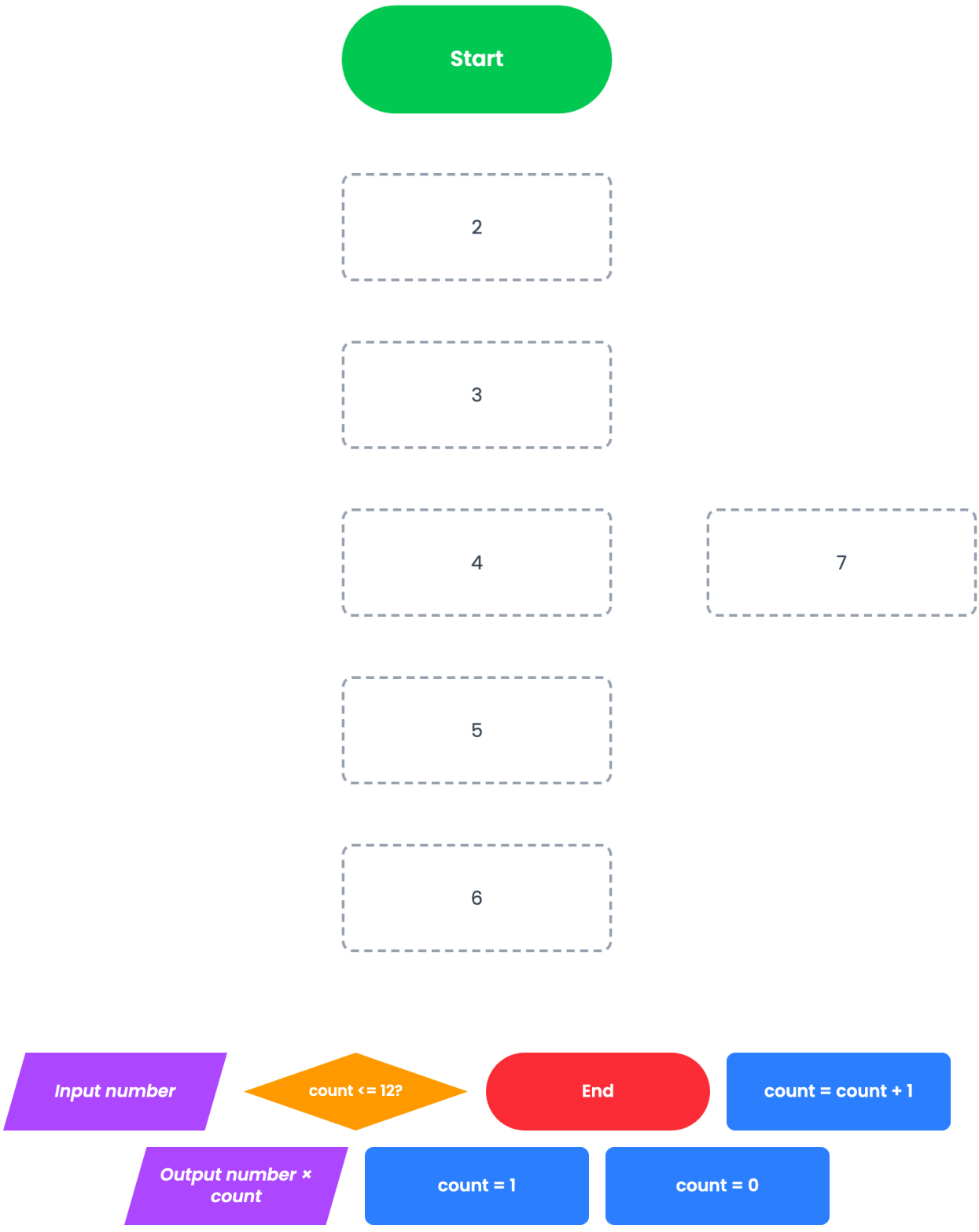


Level 18: Times Table

Medium · Draw Arrows

Input a number and output its 12 times table. Start is already placed for you.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.

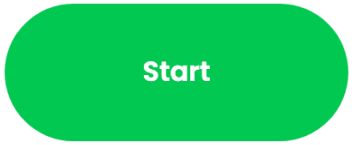


Level 19: Total Until Zero

Medium · Draw Arrows

Keep adding numbers to a total until the user enters 0, then output the total. Start is already placed for you.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.



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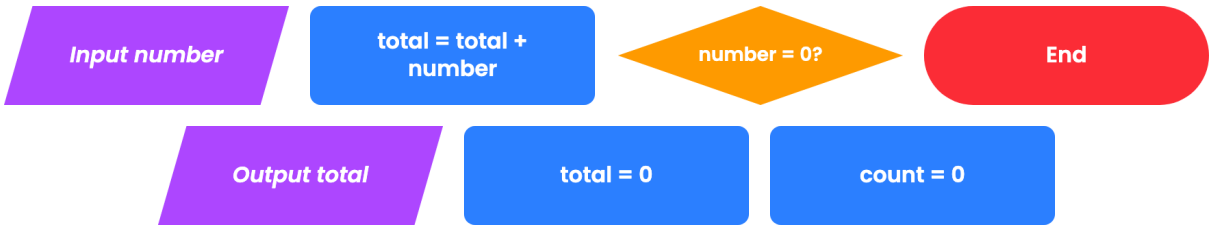
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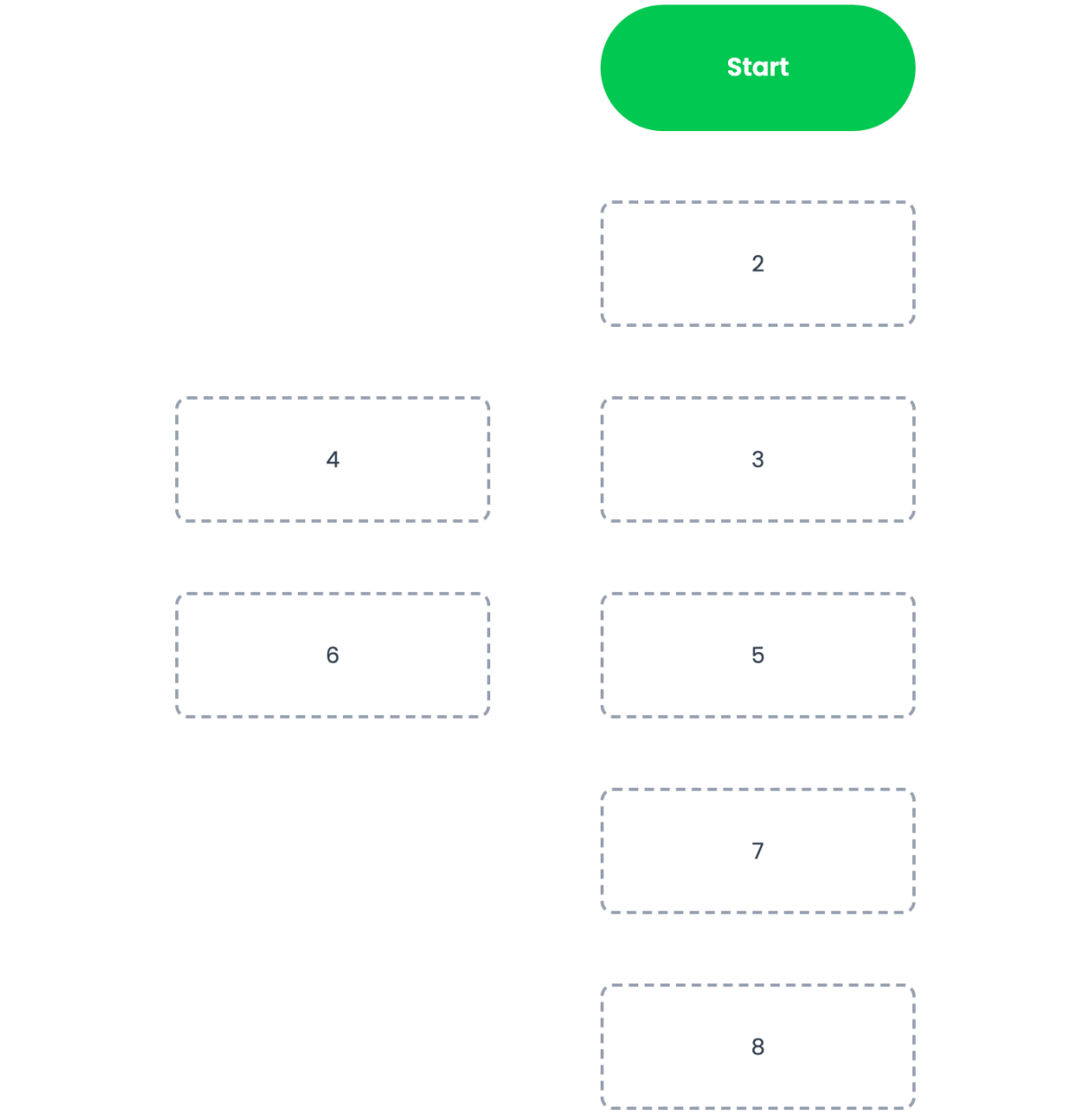


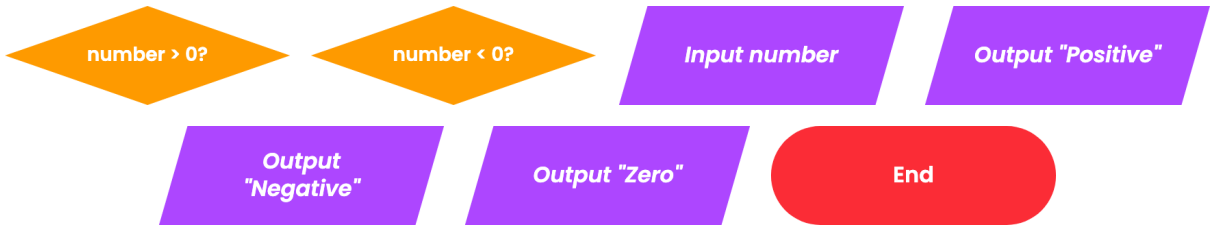
Level 20: Number Sign

Medium · Draw Arrows

Output whether a number is positive, negative or zero. Start is already placed for you.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.



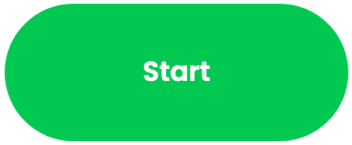


Level 21: Input Validation

Hard · Draw Arrows

Ask for a number from 1 to 10. If it is not in range, say so and ask again. Start is already placed for you.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.



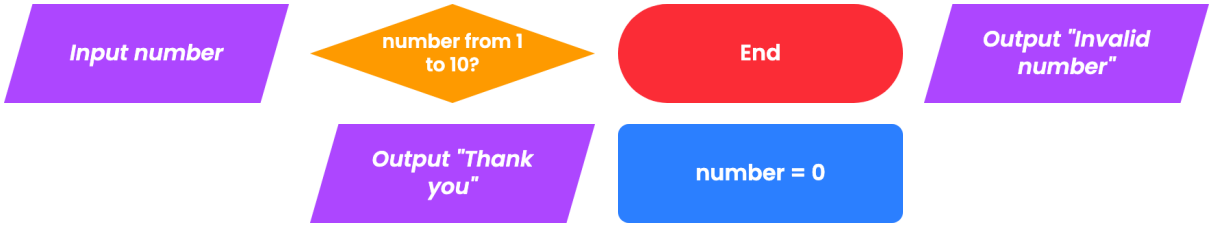
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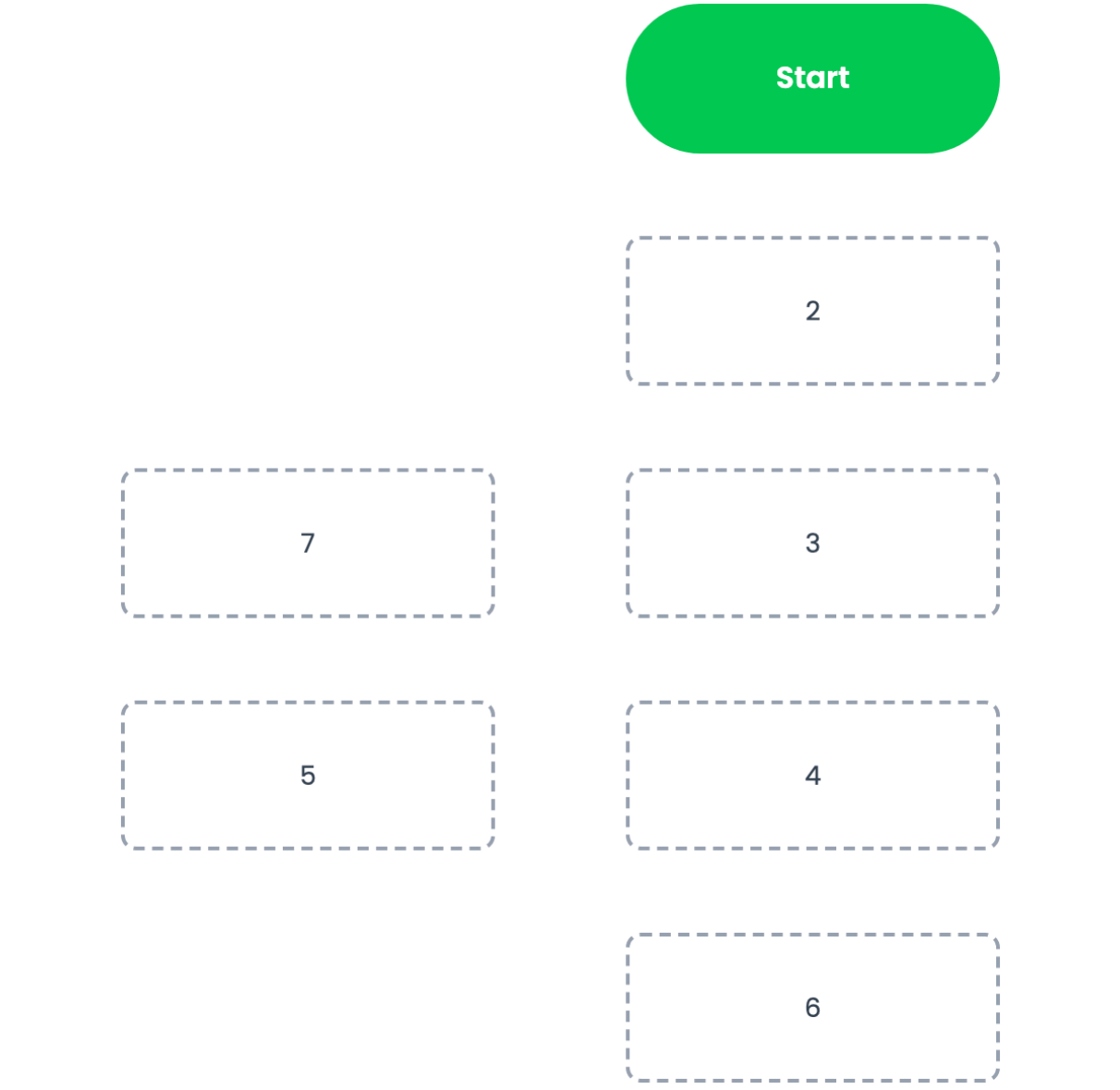


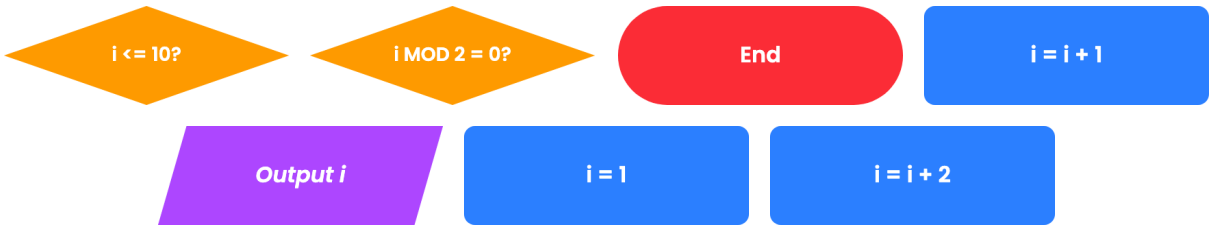
Level 22: Even Numbers

Hard · Draw Arrows

Output only the even numbers from 1 to 10. A decision sits inside the loop. Start is already placed for you.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.



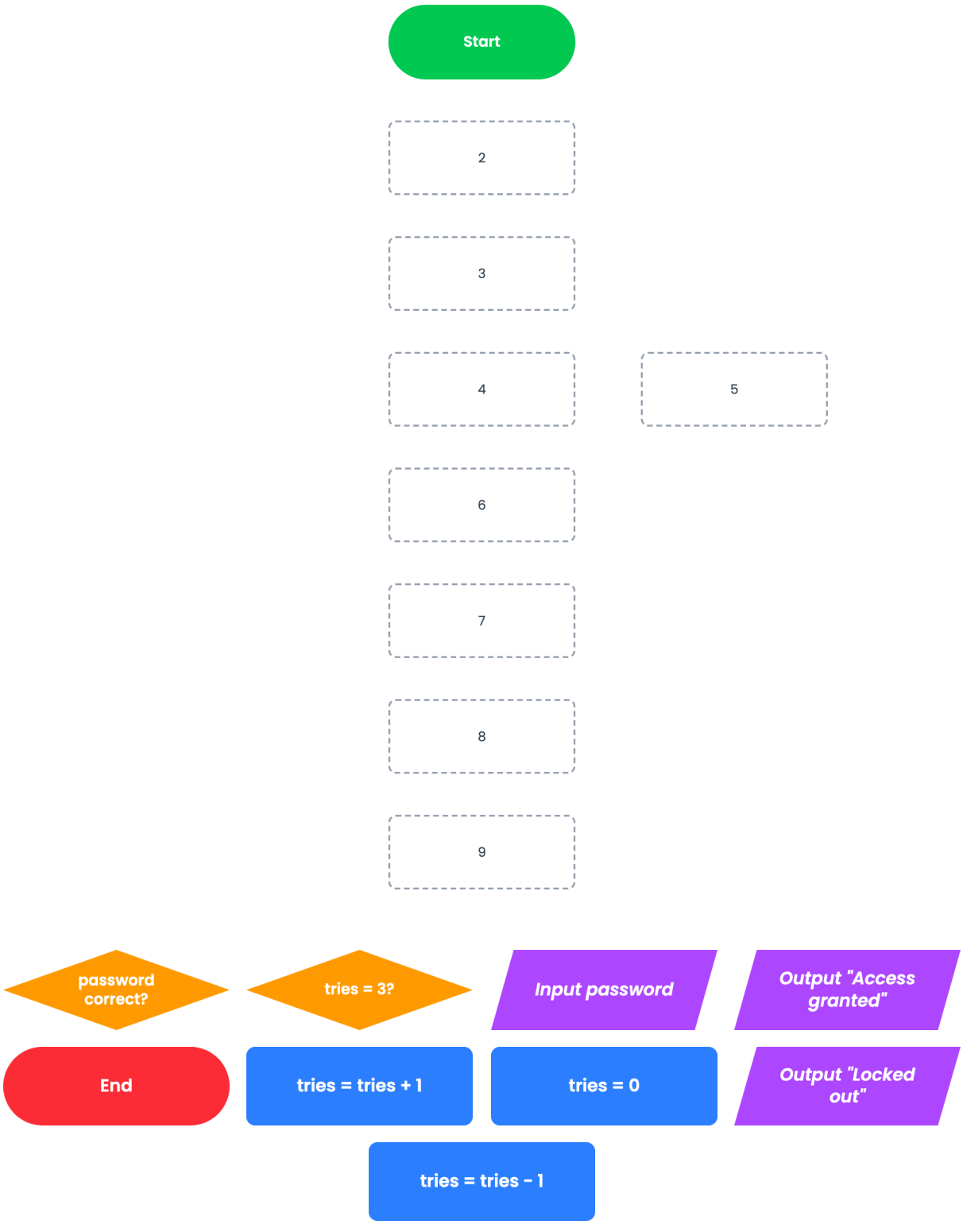


Level 23: Three Attempts

Hard · Draw Arrows

The user gets three tries to enter the correct password. After three wrong tries they are locked out. Start is already placed for you.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.

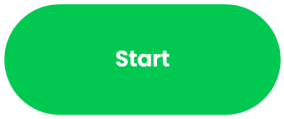


Level 24: Sum to N

Hard · Draw Arrows

Input a number n and output the total of 1 + 2 + … + n. Start is already placed for you.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.



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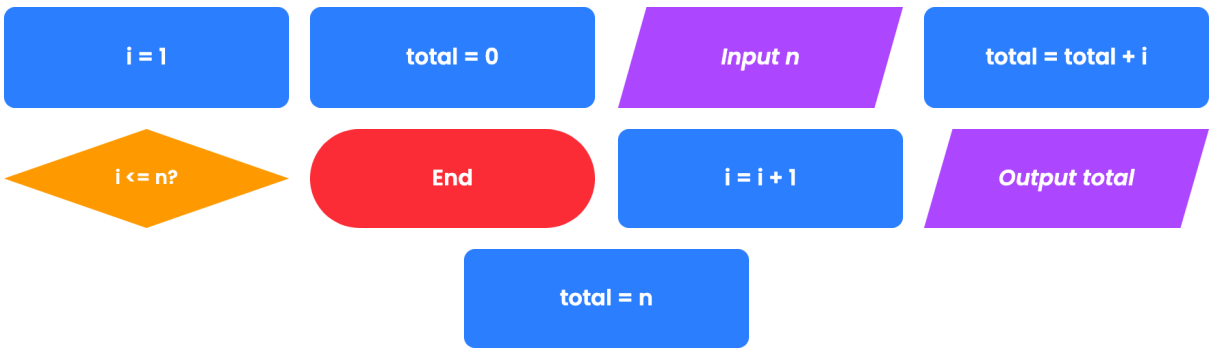
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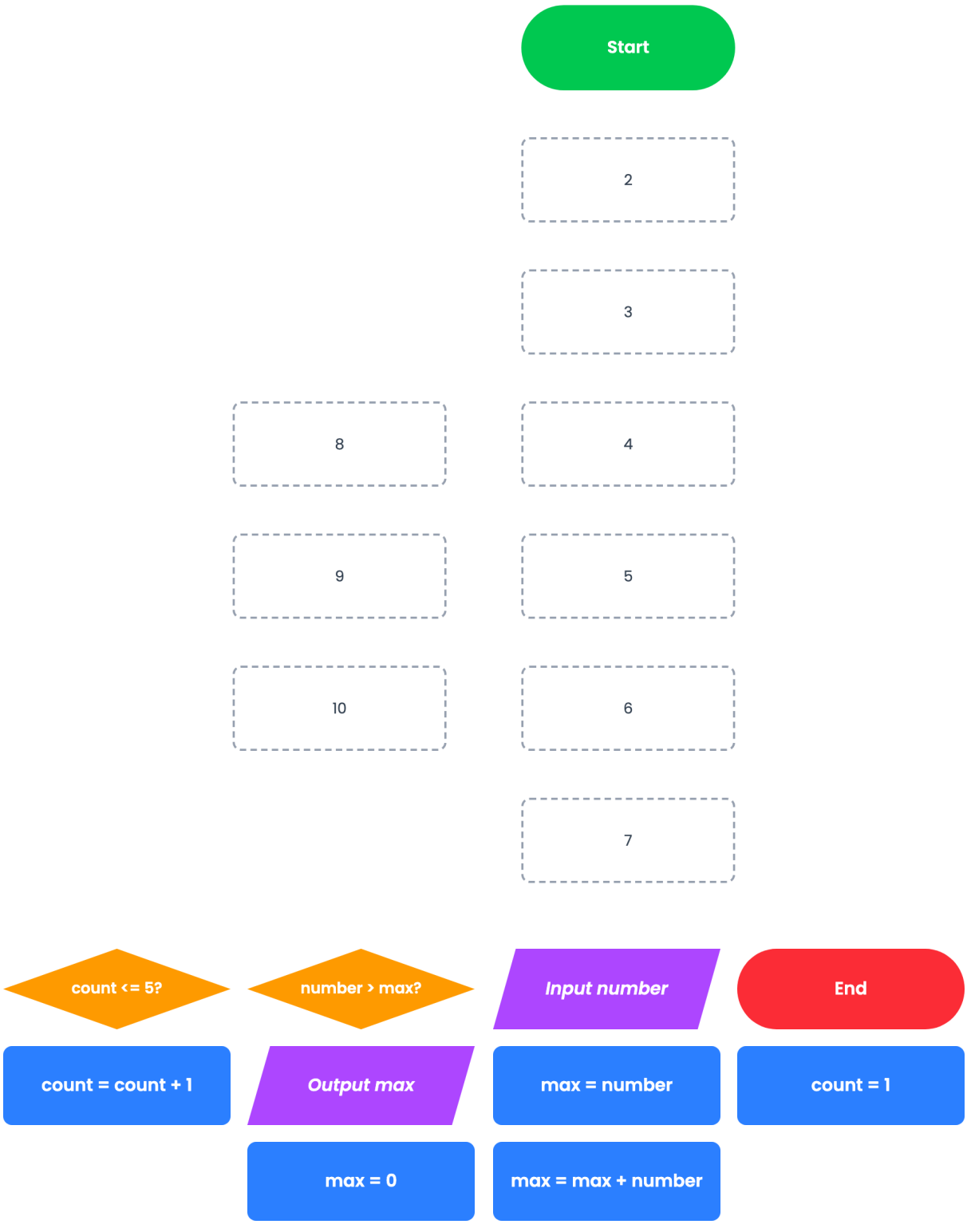


Level 25: Largest of Five

Hard · Draw Arrows

Input five numbers and output the largest. Keep track of the biggest so far. Start is already placed for you.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.

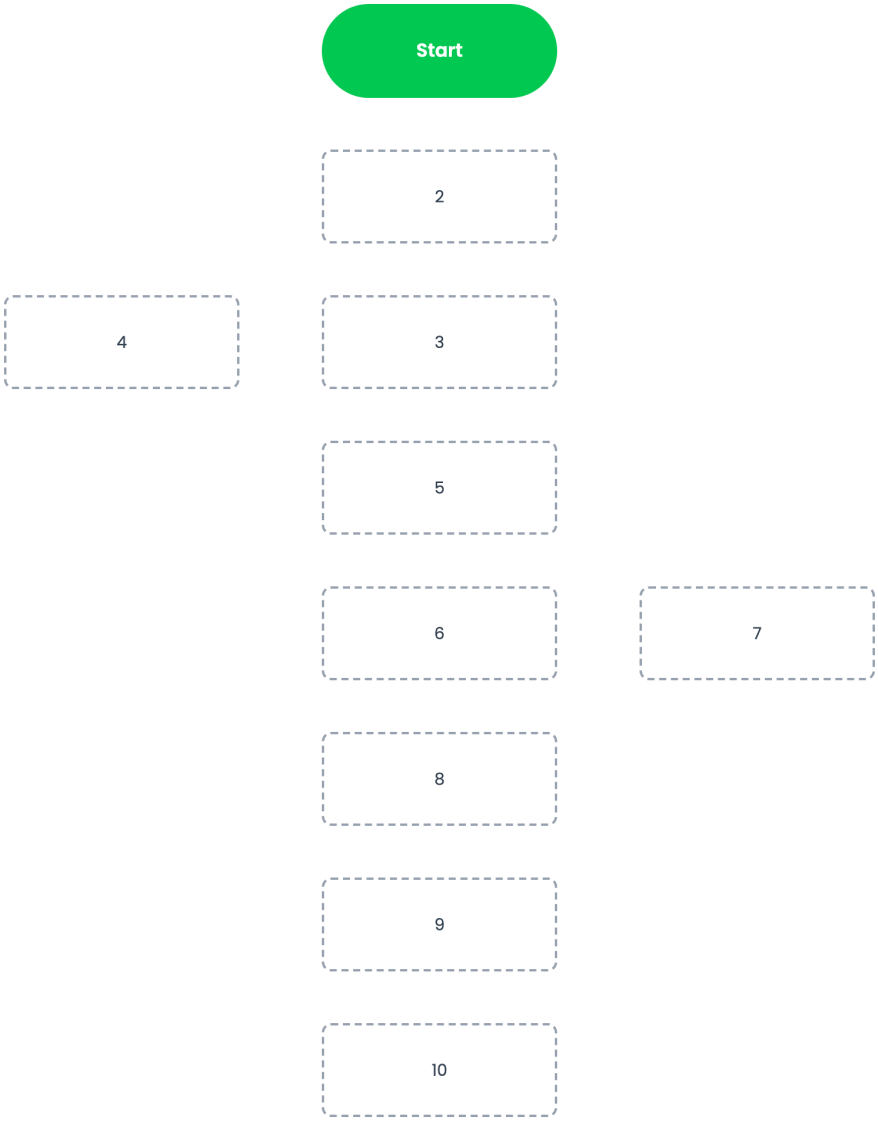


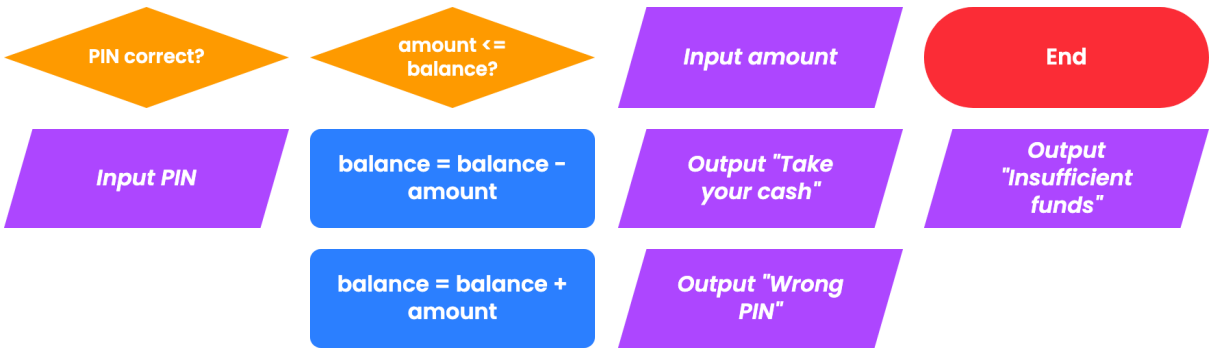
Level 26: Cash Machine

Hard · Draw Arrows

Check the PIN, then check there is enough money before paying out. Every path finishes at End. Start is already placed for you.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.



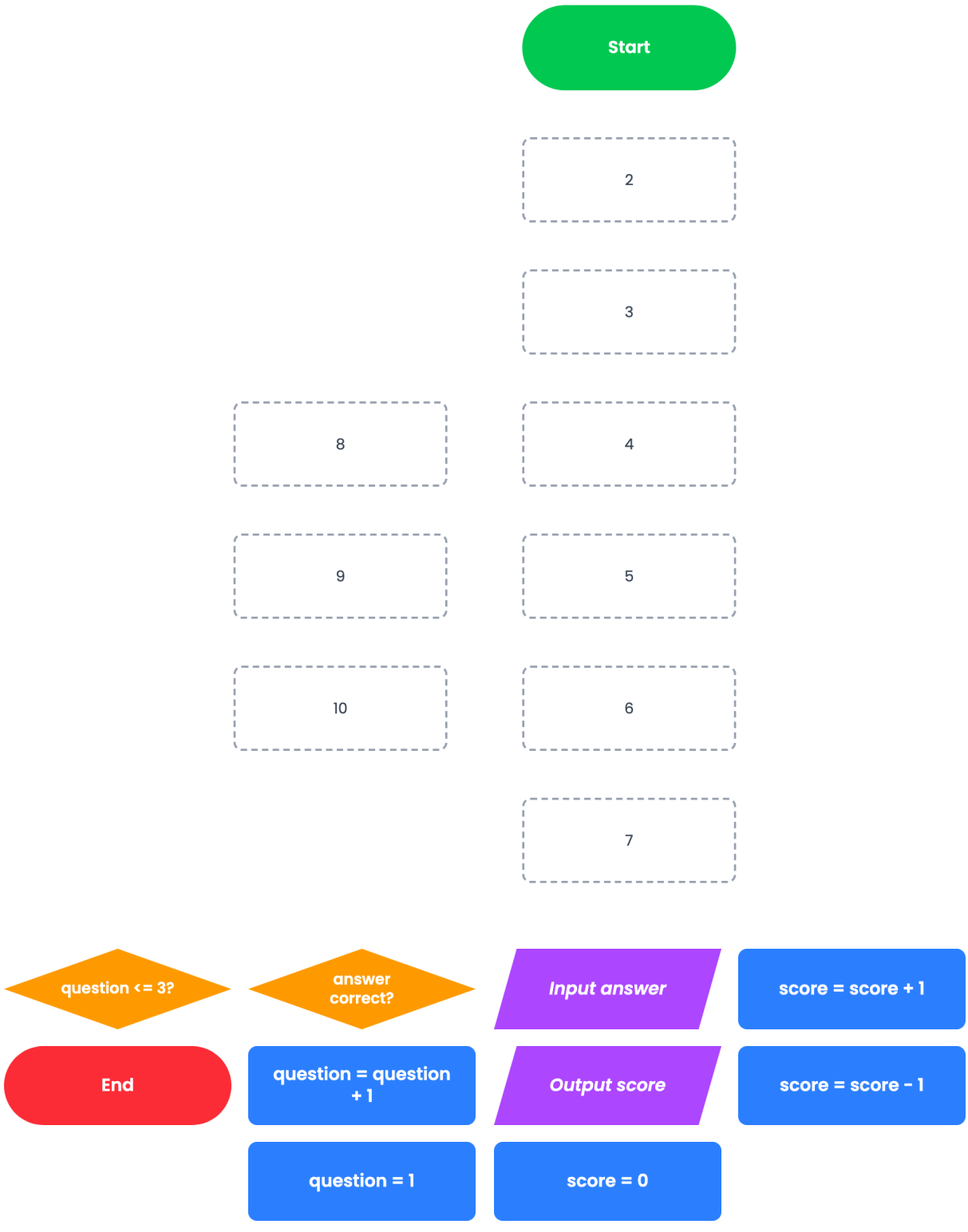


Level 27: Quiz Score

Hard · Draw Arrows

Ask three questions. Add one to the score for each correct answer, then output the score. Start is already placed for you.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.

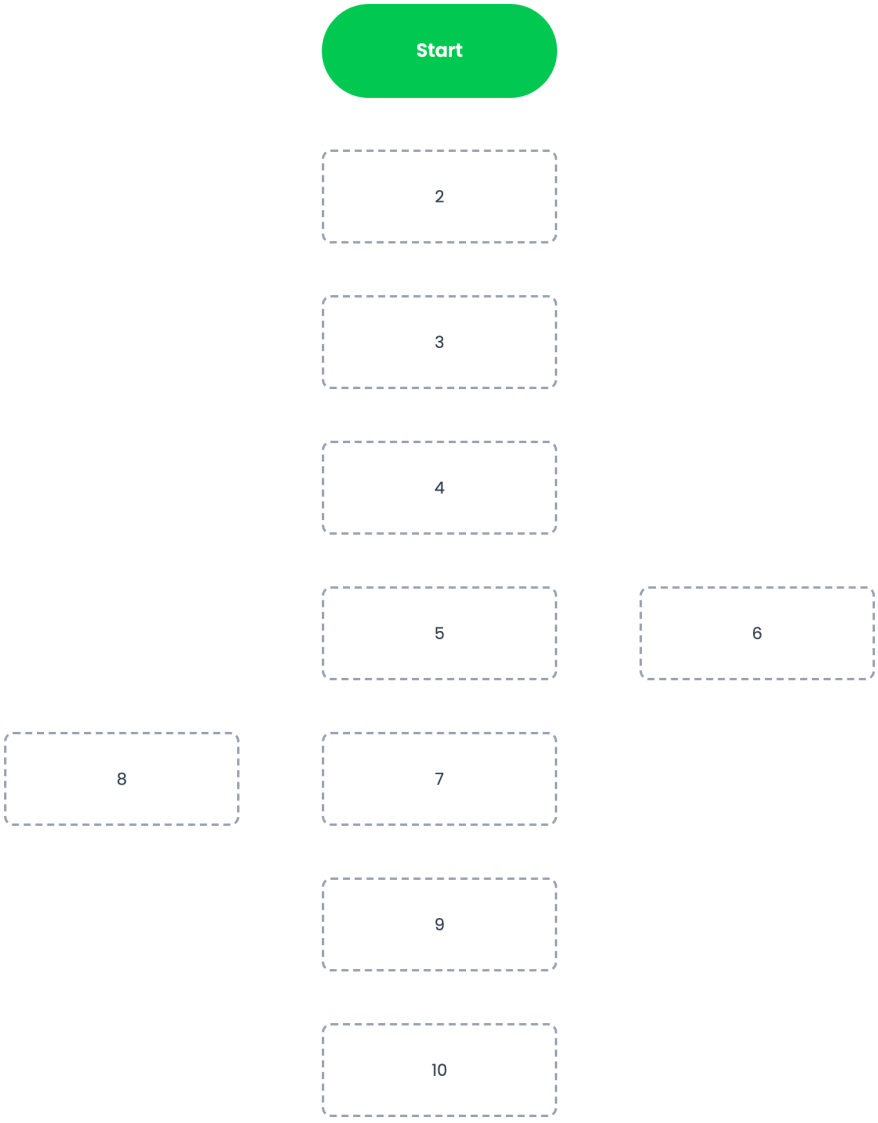


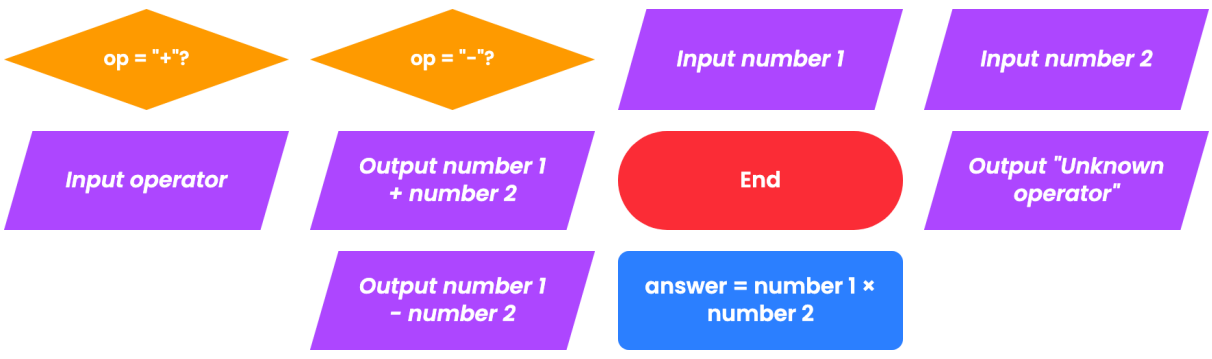
Level 28: Calculator

Hard · Draw Arrows

Input two numbers and an operator (+ or -), then output the answer. Anything else is an unknown operator. Start is already placed for you.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.



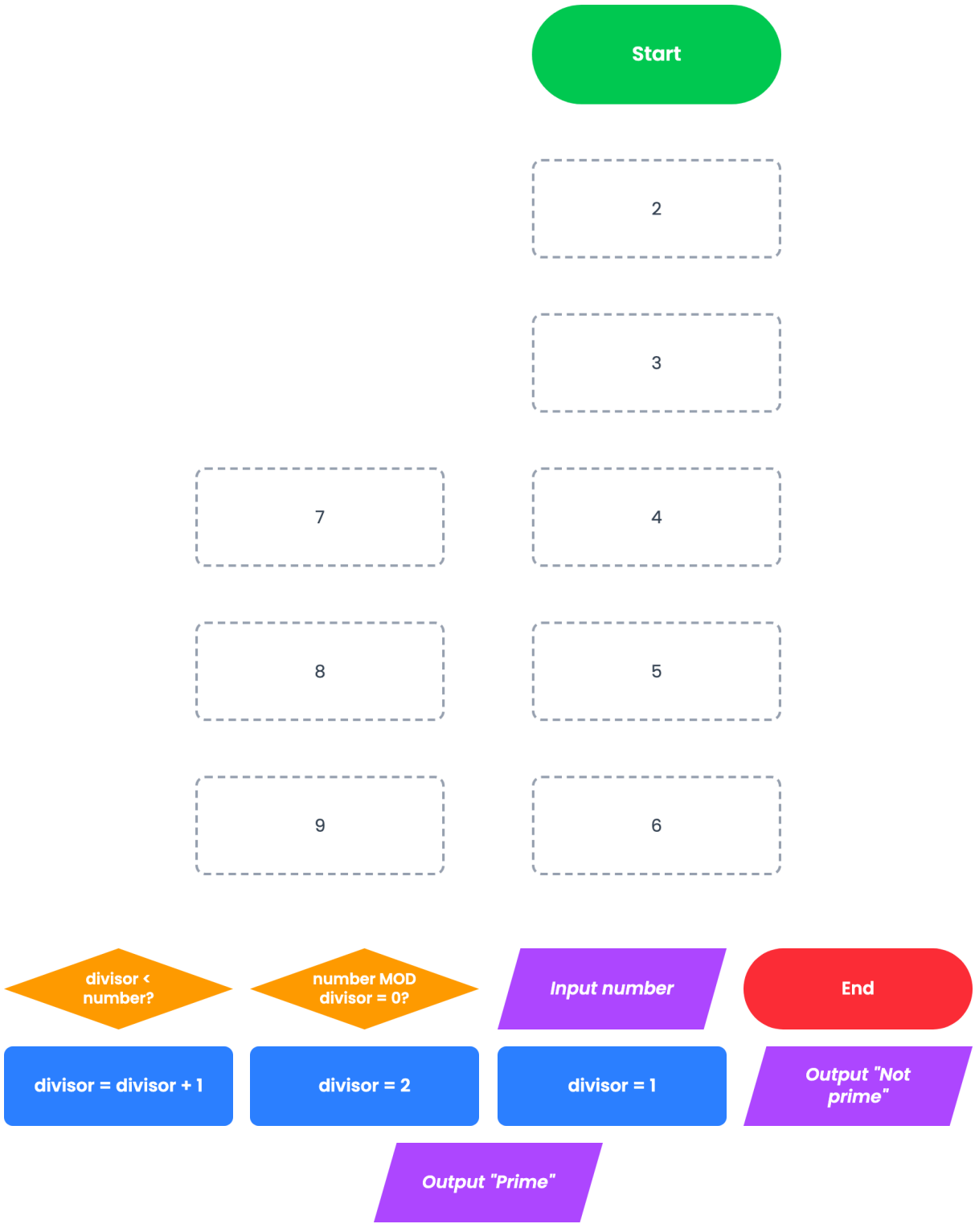


Level 29: Prime Number

Hard · Draw Arrows

Try dividing by 2, 3, 4… up to one less than the number. If any divide exactly it is not prime. Start is already placed for you.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.



Level 30: Menu Loop

Hard · Draw Arrows

Show a menu: 1 says hello, 2 says goodbye, 3 quits. Any other choice asks again. Start is already placed for you.

No arrows are drawn. Decide which symbol belongs in each numbered box and draw it in (or cut out the symbols and stick them in), then draw every arrow between them. Arrows leaving a diamond must be labelled Yes or No. Some symbols are not needed.

